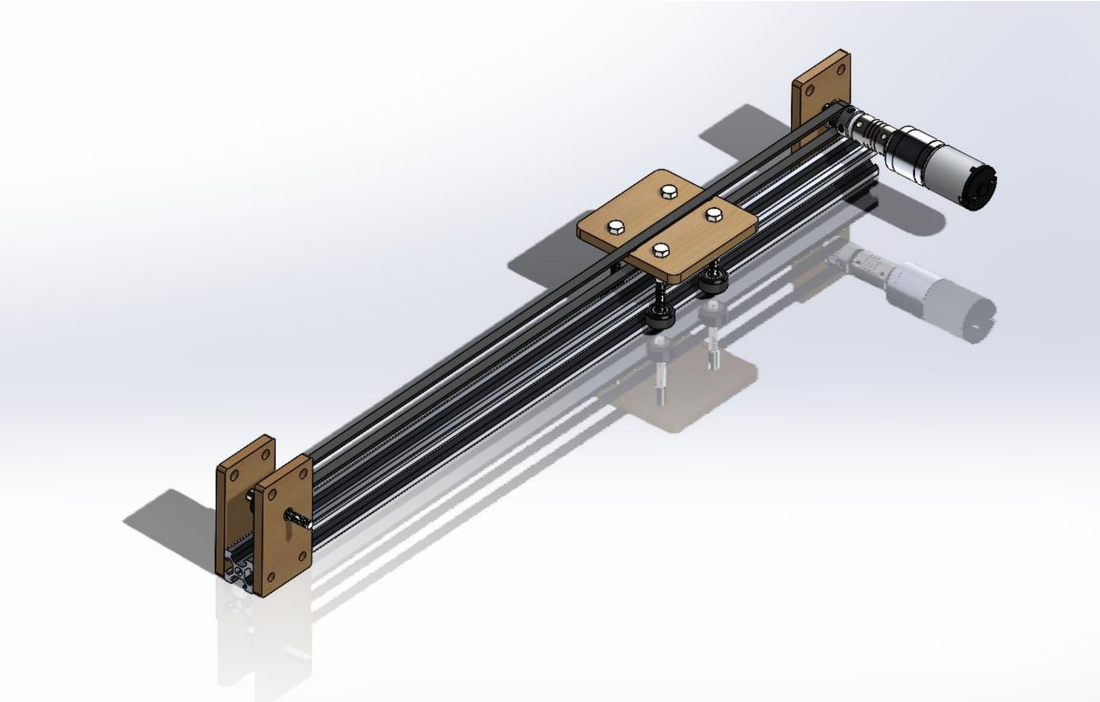
AIN SHAMS UNIVERSITY-

FACULTY OF ENGINEERING

[MCT234s]

# Modeling and Simulation of Mechatronics systems



# TEAM 8

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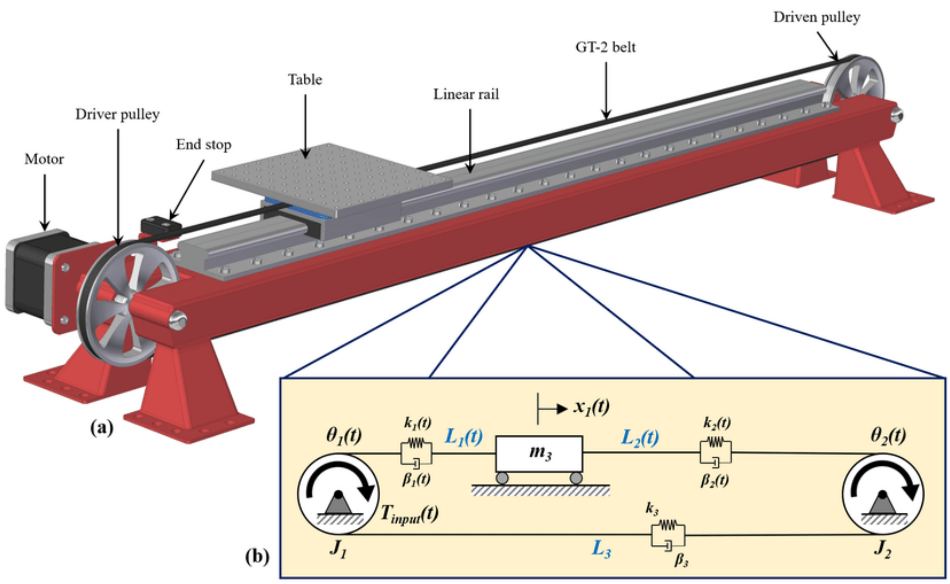
## ABSTRACT

This work presents the creation of a simulation model for a belt-and-cart system using SolidWorks, Simulink, and MATLAB. The primary goal is to estimate and identify the system's parameters accurately, including mass, damping coefficients (incorporating friction), and belt tension characteristics. The belt-and-cart system is widely utilized in mechanical applications and requires precise modelling to understand its dynamic behaviour.

The simulation model is developed by integrating a 3D mechanical design from SolidWorks with Simulink for dynamic analysis. MATLAB is employed for numerical computations and parameter extraction. The damping coefficients account for frictional effects, which are crucial for realistic modelling. The model considers key elements such as the belt's elasticity, cart inertia, and interactions between the belt and pulley system. Parameter estimation is achieved by comparing simulated outputs with experimental data or predefined system behaviours.

The proposed approach provides an efficient framework for studying the dynamics of belt-and-cart systems, enabling accurate parameter determination without requiring extensive physical testing. This model serves as a foundation for further studies, including control system design and optimization of similar mechanical systems.

## INTRODUCTION

Parameter estimation is a process in which the key parameters of a system are identified by comparing its observed behavior to a mathematical or simulated model. This technique is widely used in engineering and science to fine-tune models for greater accuracy, enabling better predictions and control of system performance. In mechanical systems, parameter estimation helps determine values such as mass, damping coefficients, and stiffness, which are often challenging to measure directly but are critical for understanding system dynamics.

Simulink, a MATLAB-based simulation environment, is an essential tool for modelling, simulating, and analysing dynamic systems. It provides a graphical interface for building models using blocks that represent mathematical functions, physical components, and signal processing operations. Simulink is widely used in academia and industry for designing and testing systems ranging from mechanical to electrical and control systems, making it ideal for studying the dynamics of a belt-and-cart system.

In this project, we focus on developing a simulation model of a belt-and-cart system using Simulink to estimate its critical parameters. The system consists of a cart driven by a belt-and-pulley mechanism, with dynamics influenced by factors such as mass, belt tension, and damping (including friction). By creating an accurate model and performing parameter estimation, we aim to replicate the system's behavior and gain deeper insights into its characteristics. This project serves as a hands-on application of parameter estimation techniques and demonstrates the power of Simulink in solving real-world engineering problems.

# Objective:

- to develop a simulation model for a belt-and-cart system using Simulink
- to estimate the system's key parameters accurately. This includes determining parameters such as cart mass, damping coefficients (accounting for friction), and belt stiffness, which are essential for predicting the system's dynamic behavior.

## The project aims to:

- Build a comprehensive simulation model in Simulink, integrating the dynamics of the belt, cart, and pulley system.
- Utilize parameter estimation techniques to align the simulation results with experimental or theoretical data, ensuring the model accurately reflects the real system.
- Validate the estimated parameters by comparing the simulated outputs with observed system behavior, demonstrating the reliability of the simulation.
- Through this approach, the project highlights the importance of combining simulation tools like Simulink with parameter estimation methodologies to analyse, design, and optimize mechatronic systems effectively

# Mathematical Model of the System:-

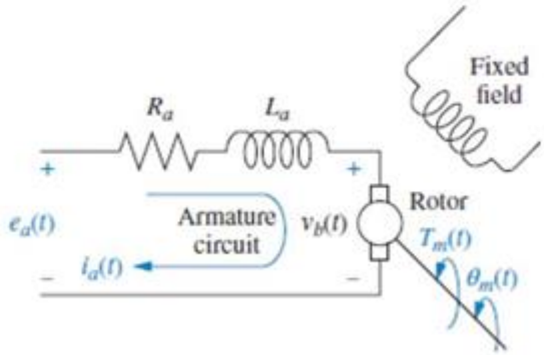



*Figure 1 DC Motor Circuit*

## 1-Electrical equation:-

$$Ea = (Ra + LS) Ia(s) + Kb\,\omega$$

$$Vin = Ea$$

## 2-Mechanical model

$$J\dot{\omega} = Tm - Tdamping - Tstatic \quad \rightarrow \text{Neglecting initial condition(static torque)}$$

$$Tm = kt \times Ia(s)$$

$$Tdamping = Beq \times \omega$$

$$JeqS\,\omega = kt \times Ia(s) - Deq \times \omega$$

Deq=Dm+Dmech.sys

Jeq=Jm+Jmech.sys

## Transfer Function:-

$$\frac{\omega}{Vin} = \frac{1}{\frac{R}{Kt}(S^2 Jeq + DeqS) + SKb}$$

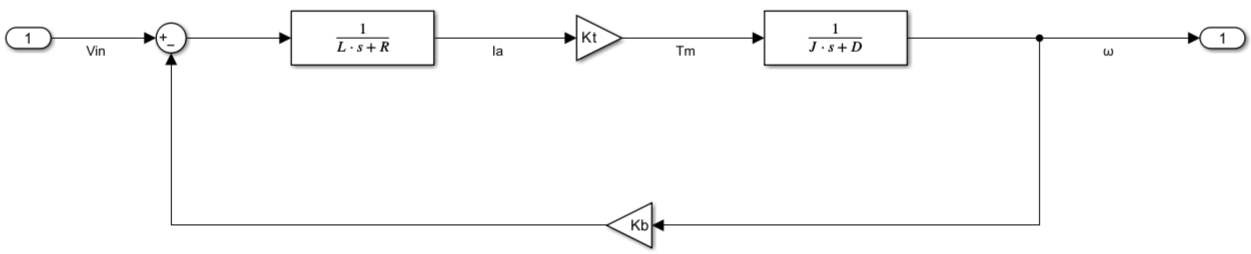



*Figure 2 Block Diagram Model Of The System*

# Readings From the System

A block diagram for Taking Reading from the real system

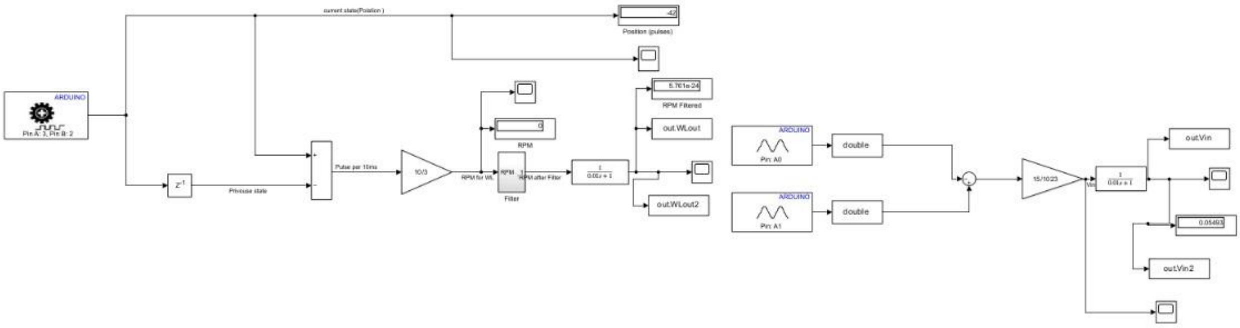


*Figure 3 Block Diagram For Motor and Encoder*

- We made a voltage divider circuit to measure the voltage across the motor through the Arduino Analog pins
- we divided the voltage across the motor by 3 to be able to measure it through the Arduino safely As max vin for pins =5 V

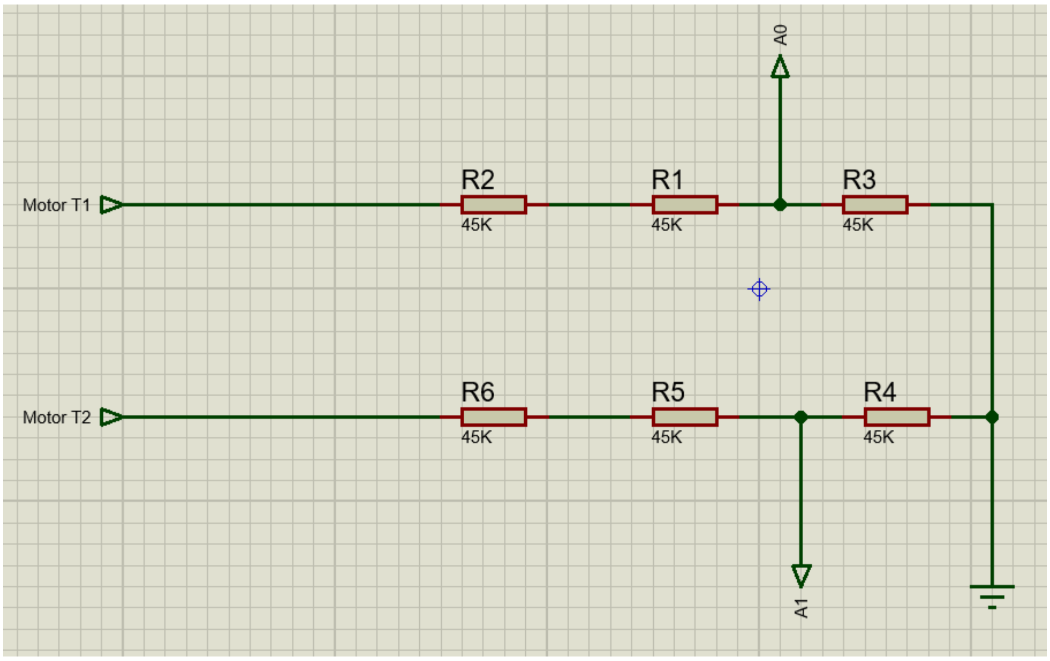



*Figure 4 Voltage Divider Circuit*

## Parameter Estimation Procedures : -

-We need to make a parameter estimation for R,Kt,Kb,L,D and J

-the system is 2DOF so we could Estimate 3 Parameter in 1 Experiment

So we separate the Parameter estimation to 2 Steps :-

### 1-Motor Estimation

To estimate R , L , Kb,Kt ,D And J of the Motor

Considering the energy conversion

$$Pm = Tm\ \omega = Pe = e\ ia \rightarrow kT\ ia\ \omega = ke\ \omega\ ia$$

$$kT = ke = k$$

that's why we assume that kT=ke=k

- we take a 4 Readings from our system to estimate the R and K we solve it and it gives us R and K with an Error 3.64%

  From Equation Vin = $\mathrm{RI}+\mathrm{L}\frac{di}{dt}+\mathrm{k}\omega$

At steady state Vin = $\mathrm{RI}+\mathrm{k}\omega$

| K | R | Vin | current | N | $\omega$ | R I +k$\omega$ | RI+k$\omega$-Vin | $(\mathrm{RI}+k\omega-\mathrm{Vin})^2$ | ((RI+k$\omega$-Vin)/Vin)x100 |
|---|---|---|---|---|---|---|---|---|---|
| 0.499 | 4.8 | 10.3 | 0.1275 | 186 | 19.47787 | 10.35169 | 0.051691 | 0.002672 | 0.501857 |
| | | 8 | 0.115 | 142 | 14.87021 | 7.98779 | -0.01221 | 0.000149 | -0.15262 |
| | | 6 | 0.097 | 104 | 10.89085 | 5.911612 | -0.08839 | 0.007812 | -1.47314 |
| | | 1.5 | 0.062 | 24 | 2.513274 | 1.554623 | 0.054623 | 0.002984 | 3.64153 |
| | | | | | | | Sum | 0.013617 | |

*Figure 5 Estimation for R and K*

- From this Estimation we take R=4.8 ohm and K=0.5

- We build motor Model at figure 16 To estimate L,D,J

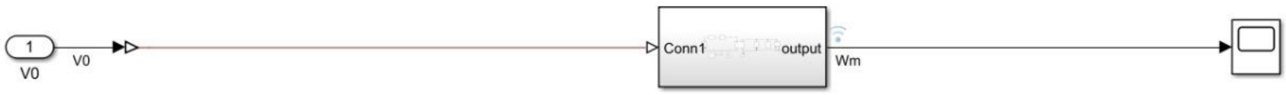



*Figure 6 Motor Model Subsystem*

From the Estimation with 10 iterations and validation we got that

Dm= 0.0044698 N.m/(rad/s)

L=4.38 mH

Jm=0.0081555 $Kg.m^2$

# Parameter Estimation for motor:-

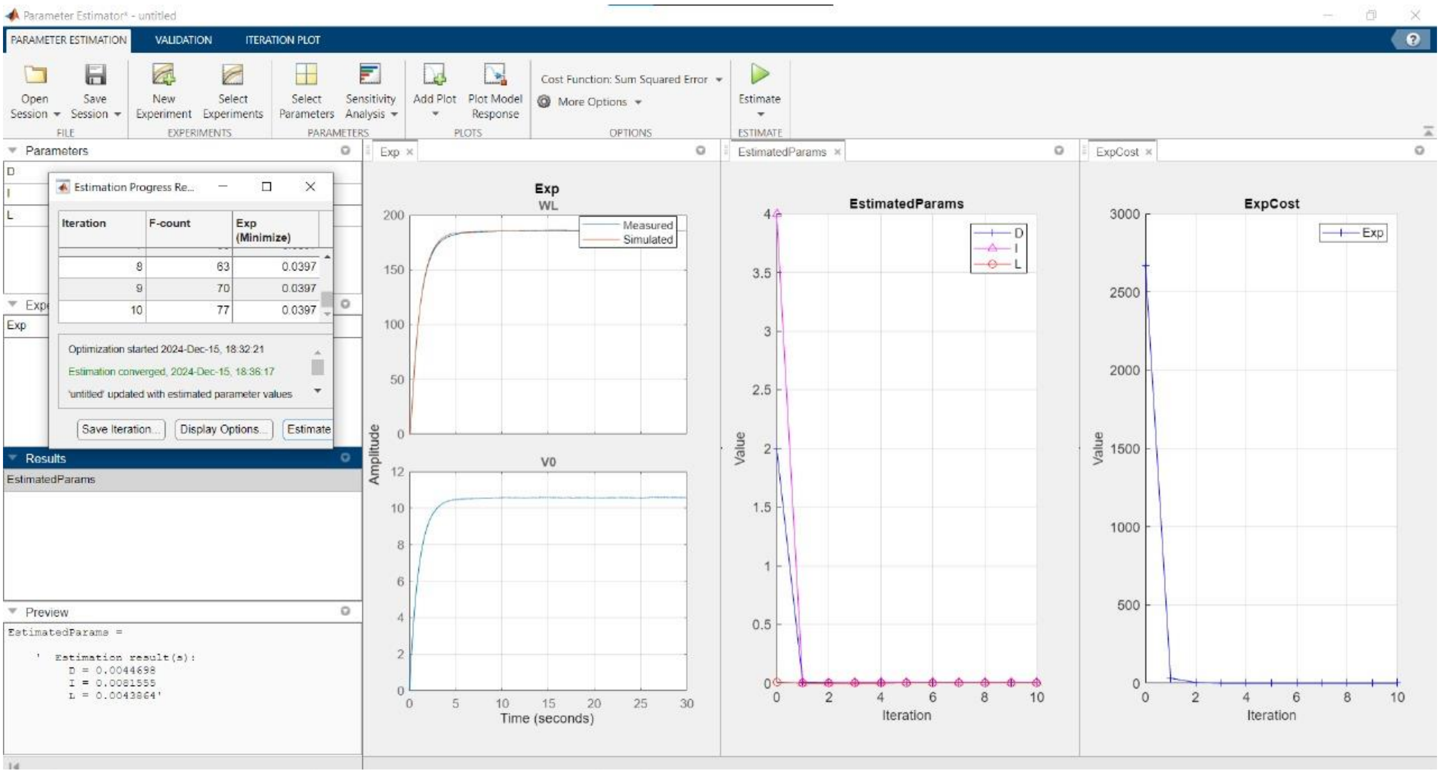


*Figure 7 Parameter Estimation Graphs*

# Validation

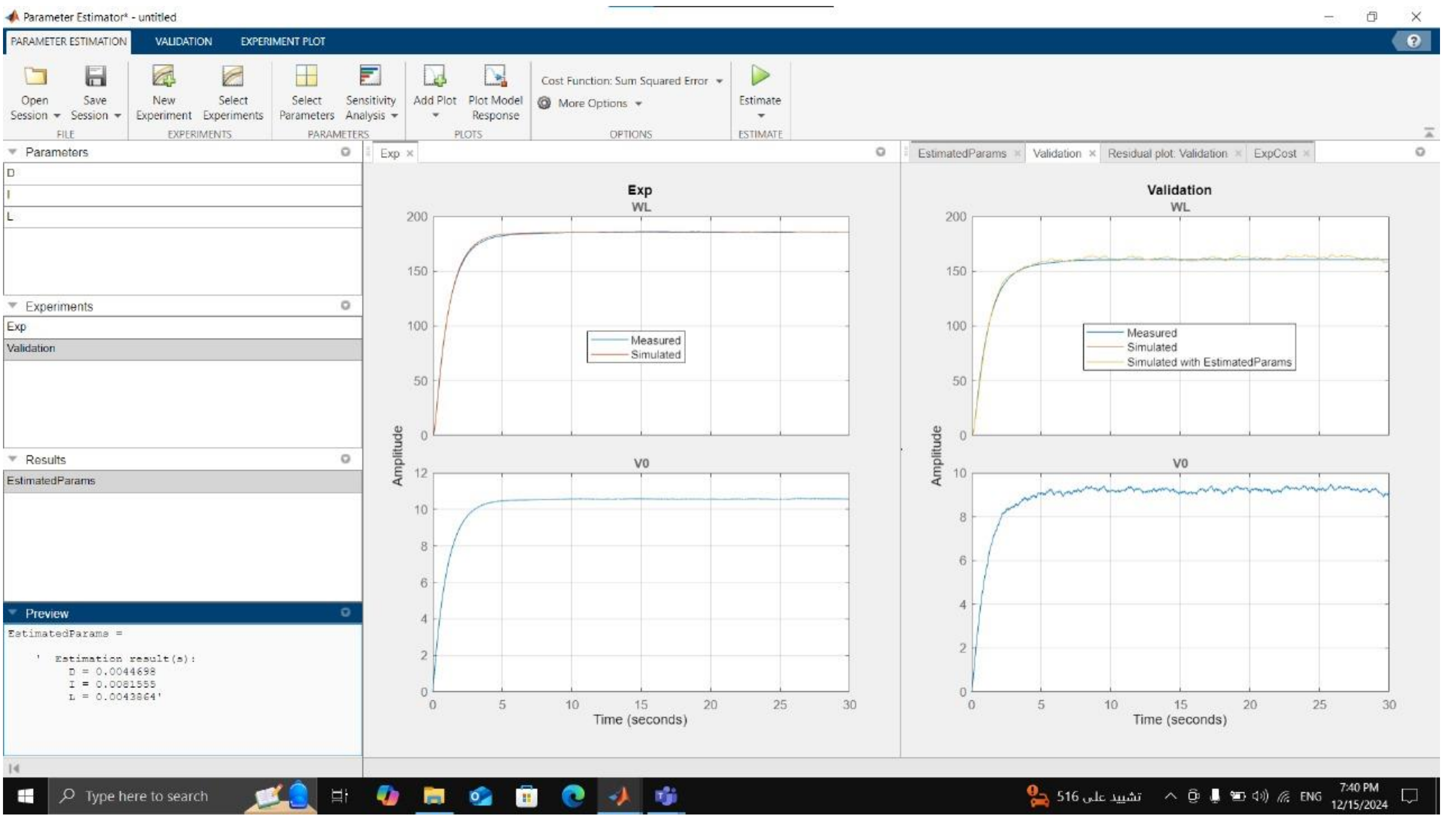


*Figure 8 Validation Graphs*

## 2- System Estimation

To estimate D and J for the mechanical system

We build motor Model at figure(17) To estimate D and J

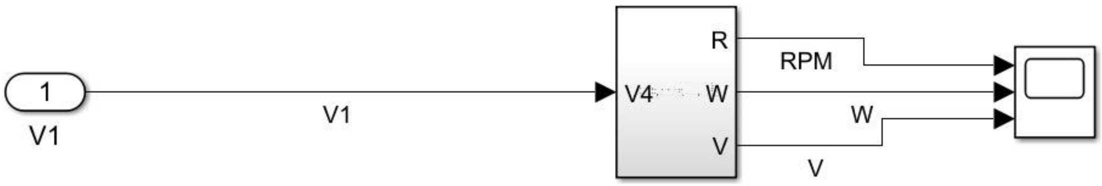



*Figure 9 System Model*

From the Estimation with 10 iterations and validation we got that

D= 0.00442798 N.m/(rad/s)

J=1.0534$\times 10^{-5}$ Kg.m$^2$

## Parameter Estimation for System:-

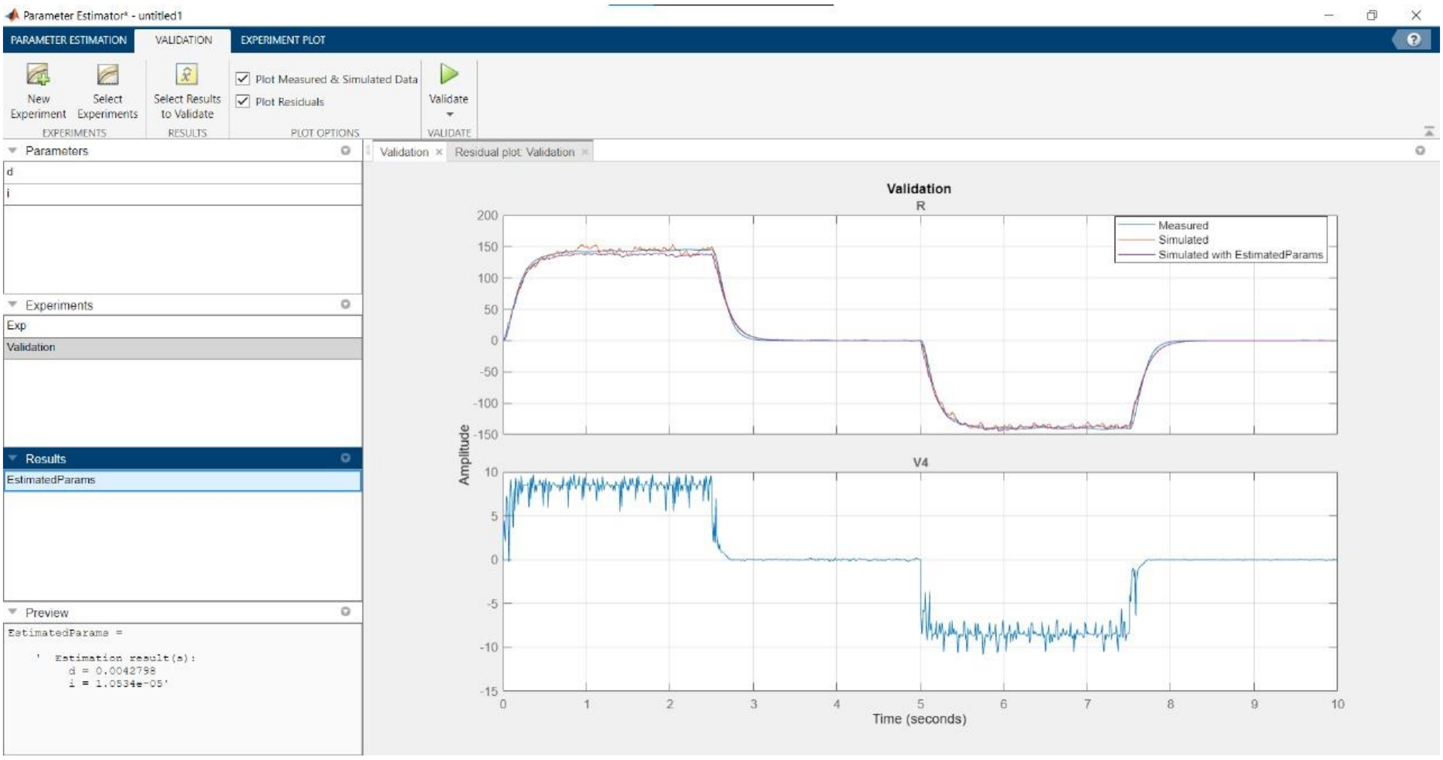


*Figure 10 Validation Graphs*

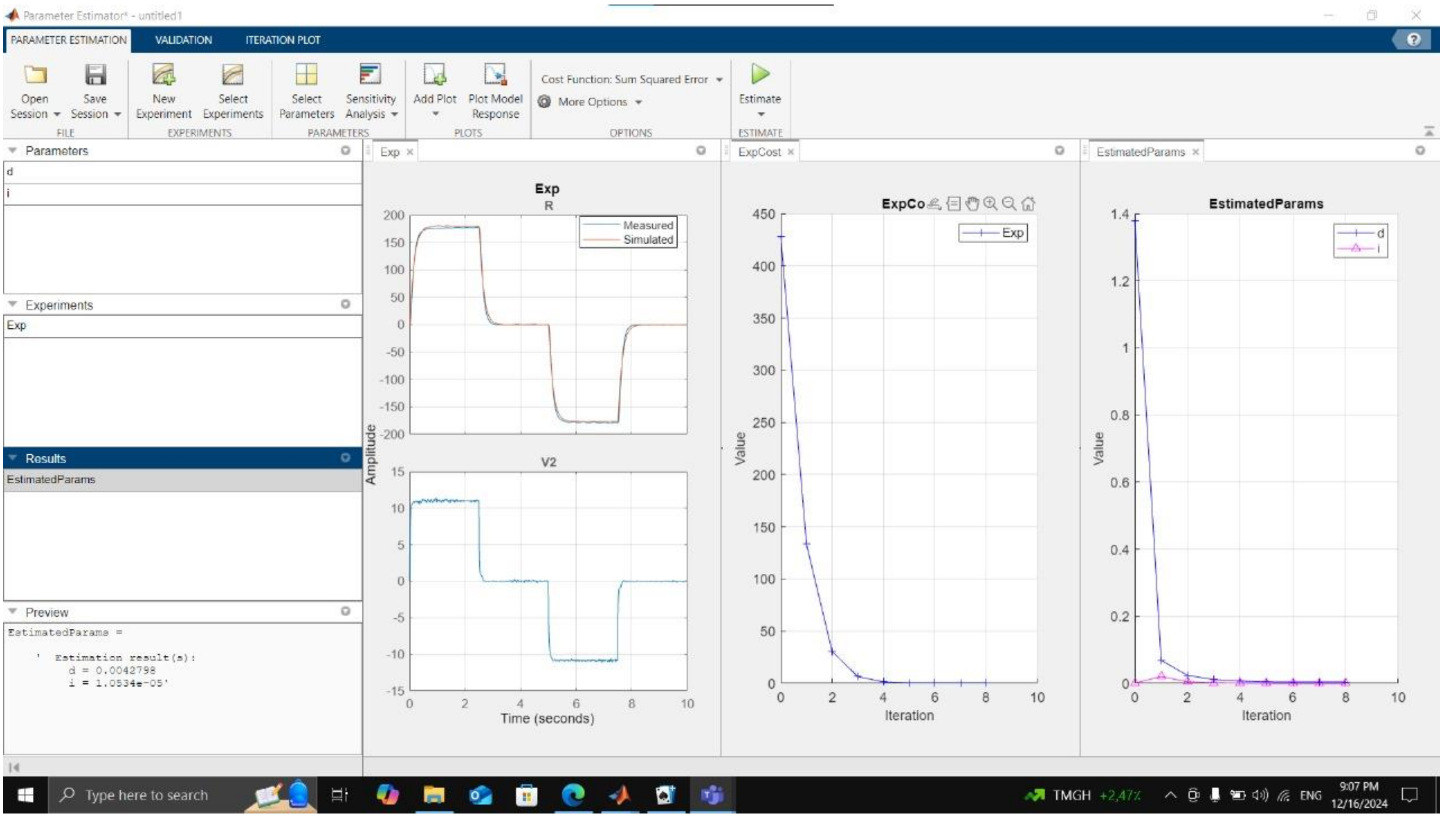


*Figure 11 Parameter Estimation graphs*

# MATLAB Block Diagram:-

## 1)For Readings:-

### 1-Encoder

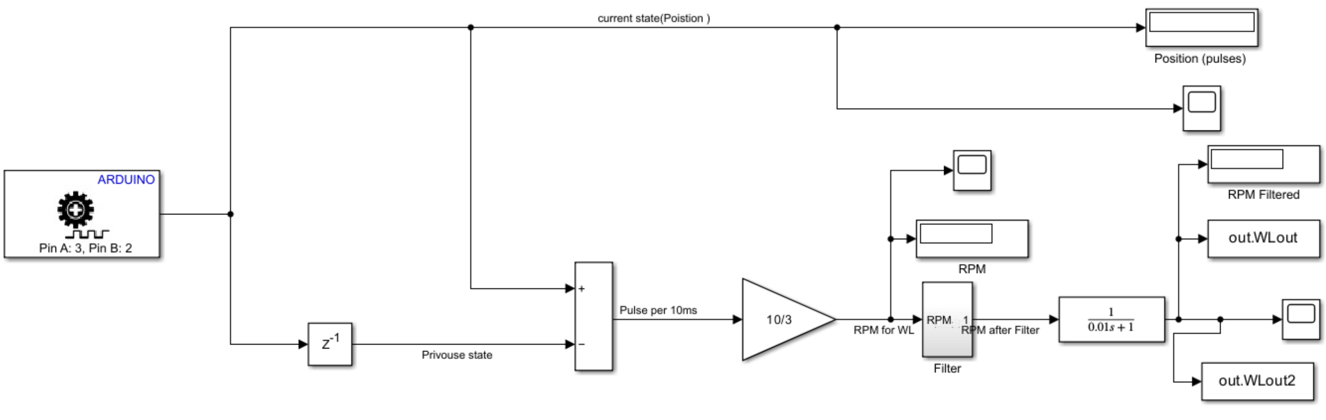



*Figure 12 Encoder Block diagram*

## *RPM Filter*

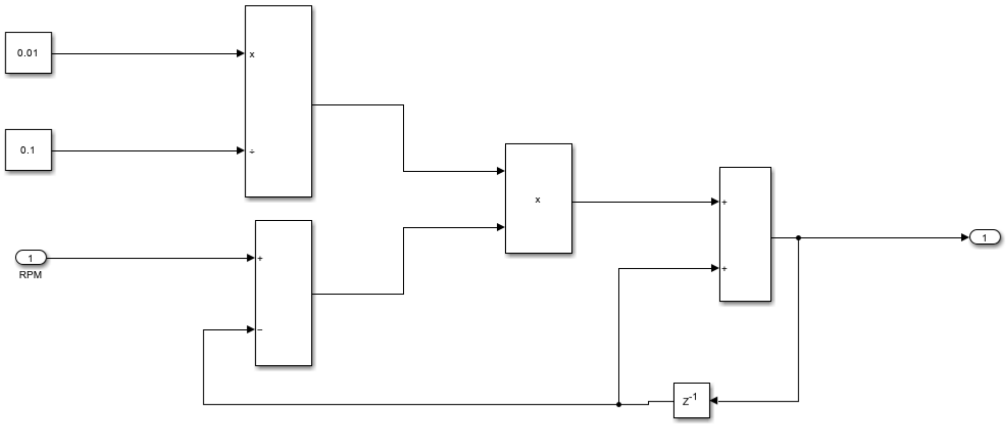



*Figure 13 Filter for Encoder Block diagram*

## 2-Motor Vin:-

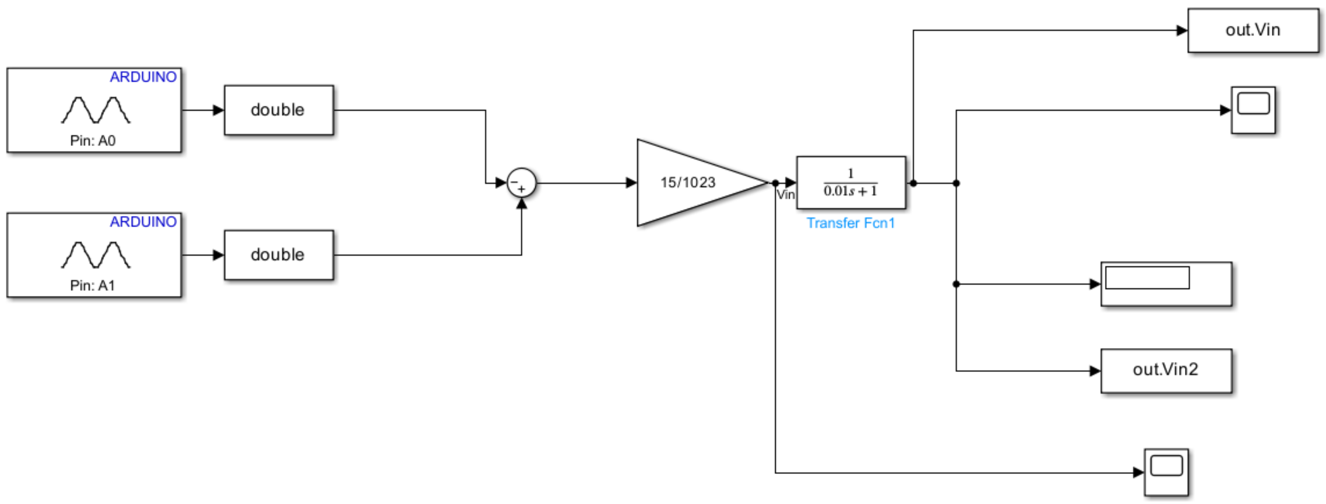



*Figure 14 Vin Block Diagram*

## 3-Control Direction

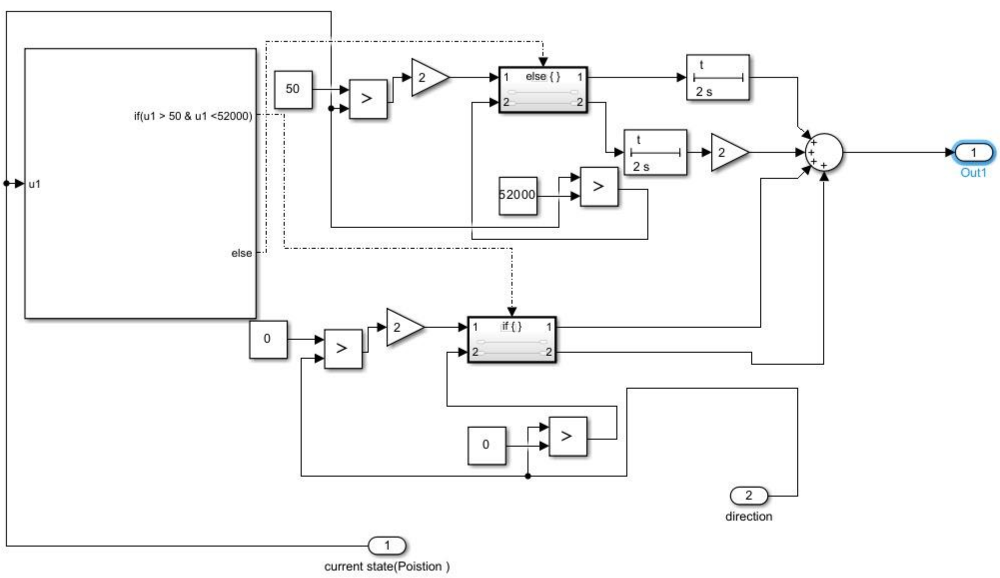



*Figure 15 Direction Control Circuit*

## 2)For Parameter Estimation :

### Motor Model:-

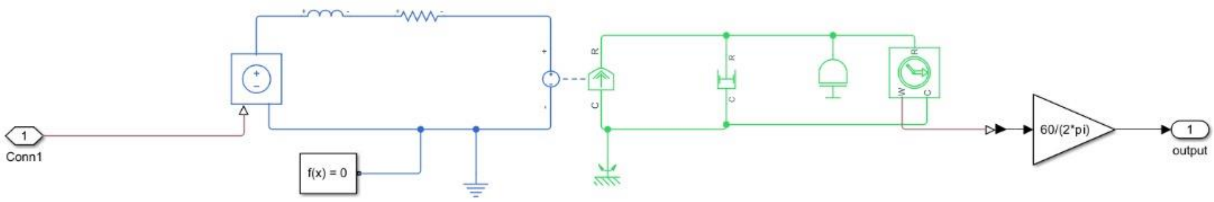



*Figure 16 Motor Model*

### System Model:-

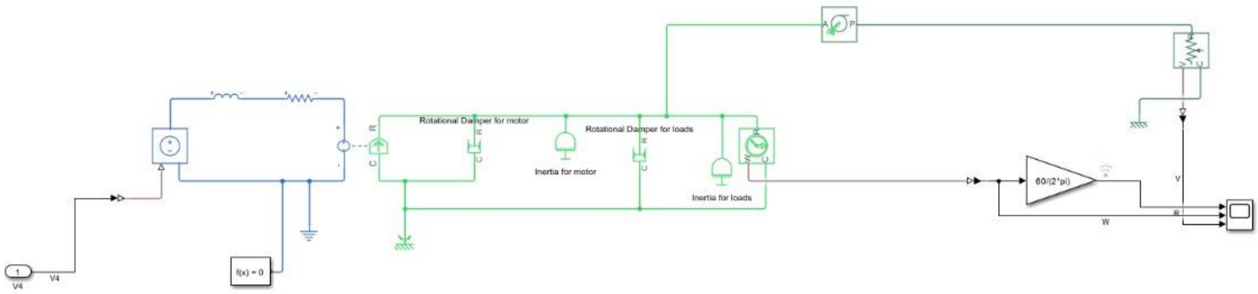



*Figure 17 System Model*

## schematic graphical:-

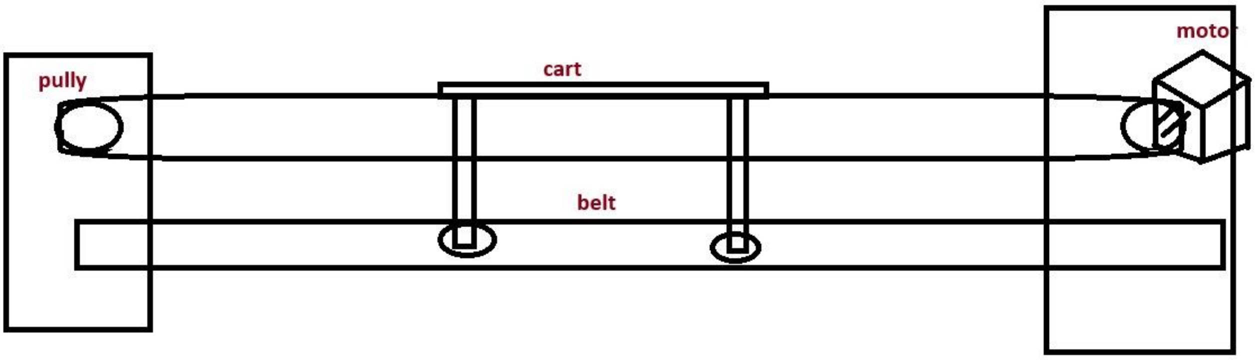



*Figure 18 initial schematic*

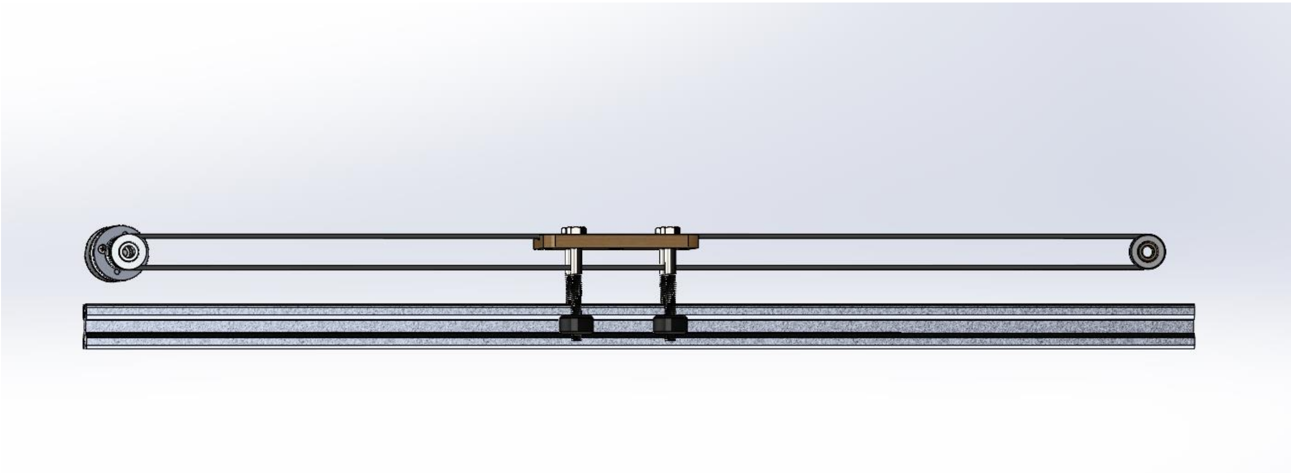


*Figure 19 initial Cad Design*

# Cad  Design:-

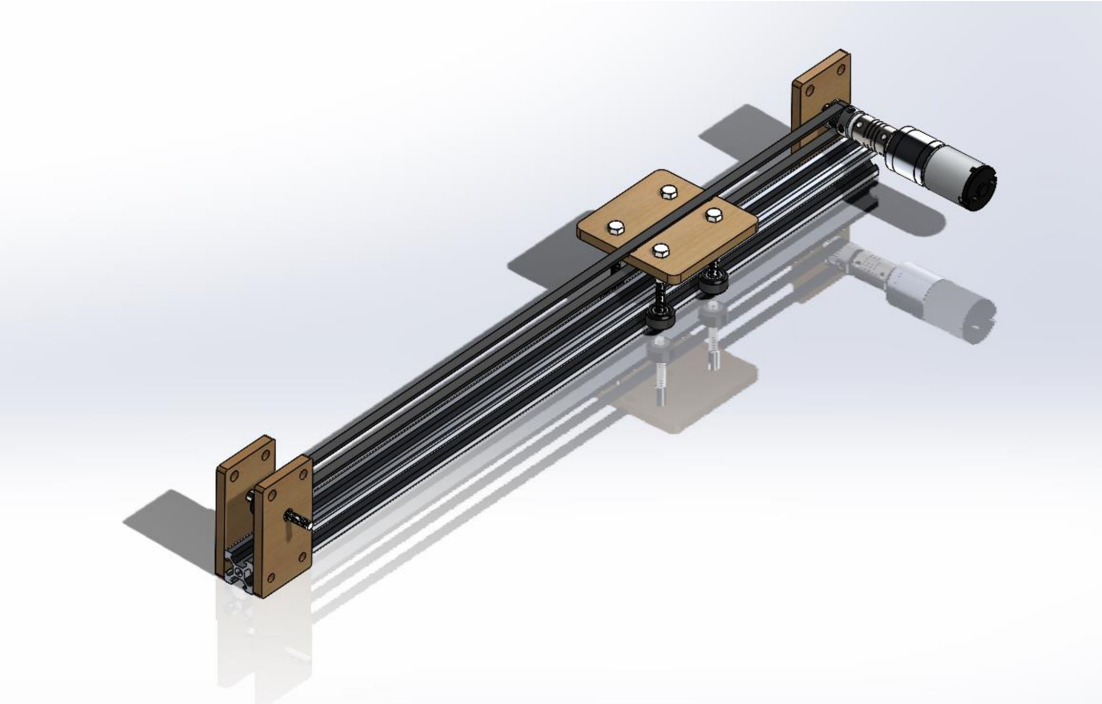


*Figure 20 CAD Design*

# Hardware Implementation: -

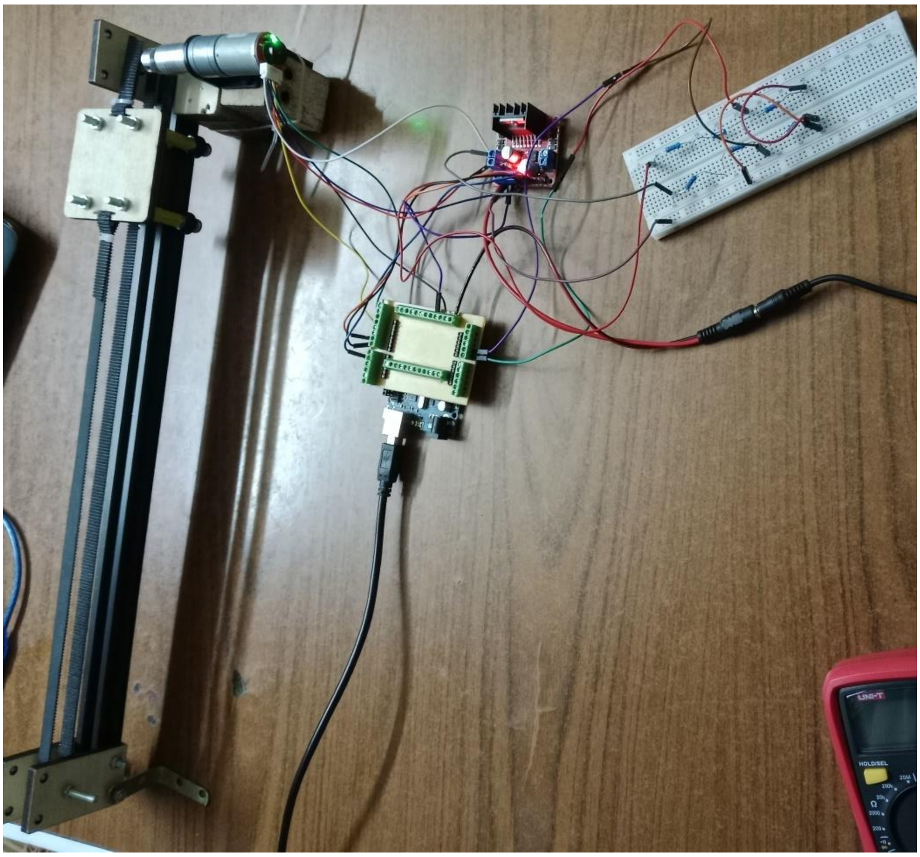


*Figure 21 Hardware Implementation*

# Simscape Multibody

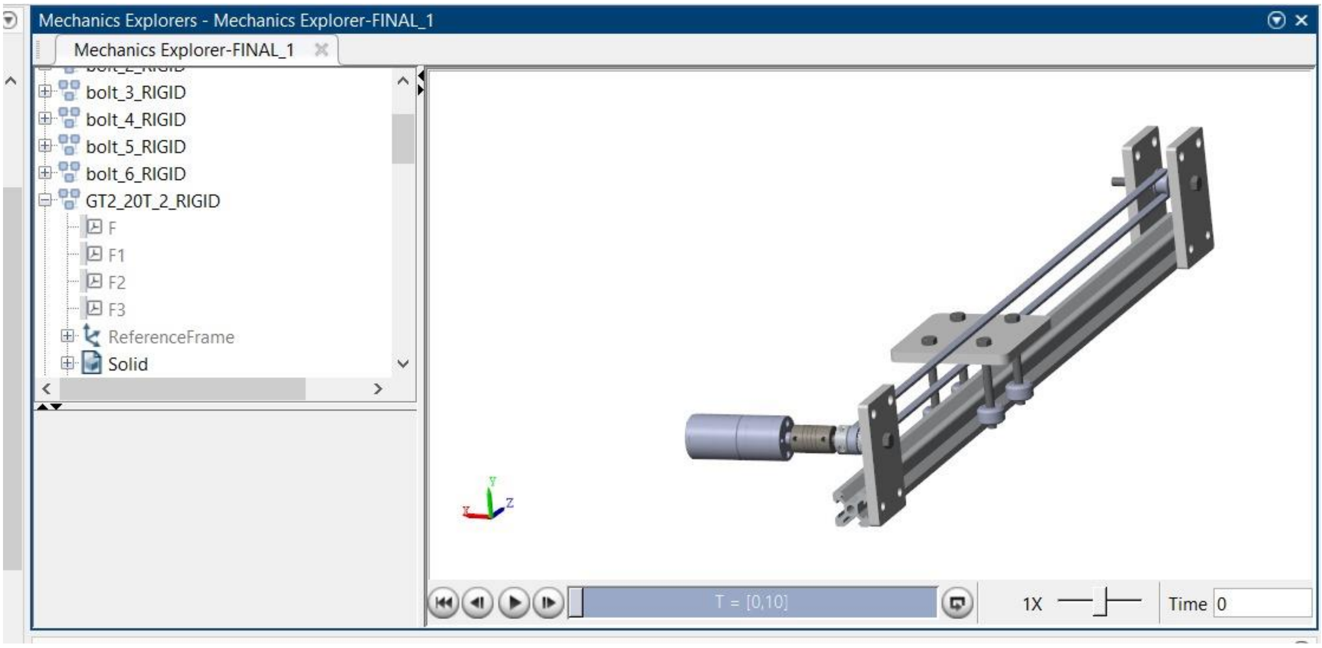



*Figure 22 Multibody Simscape Simulation*

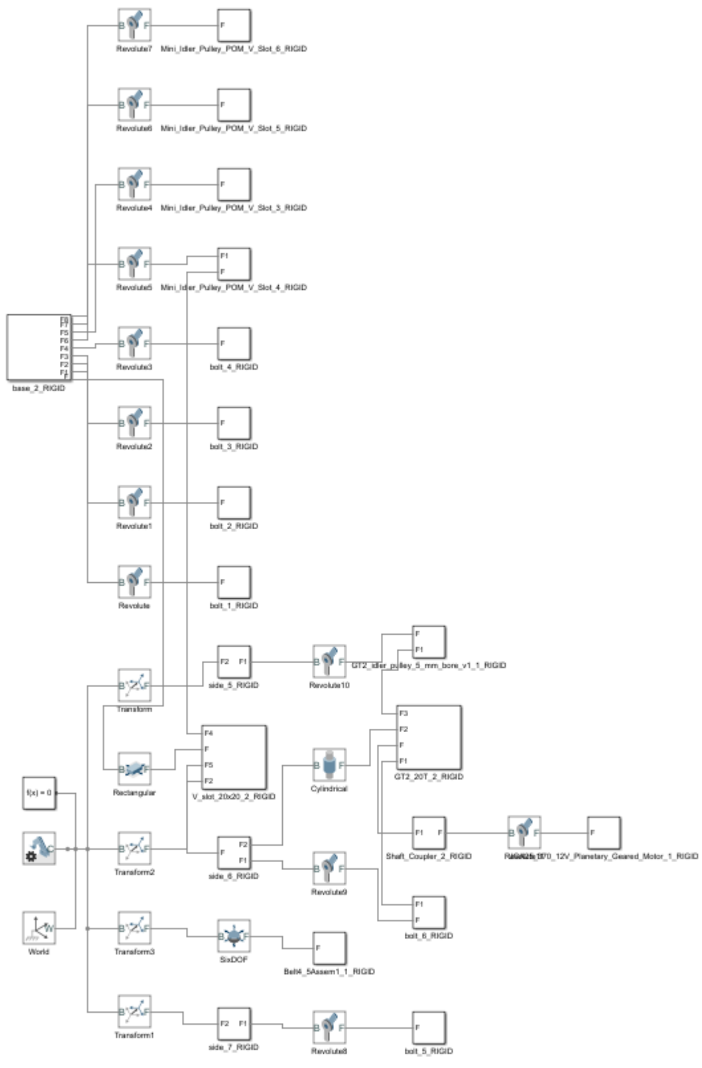



*Figure 23  Multibody Simscape Blocks*

## Conclusion:-

From Parameter estimation

Jeq=$1.0534\times 10^{-5}$+0.0081555=$8.166034\times 10^{-3}$ $Kg.m^2$

Deq= 0.00442798+0.0044698 = $8.89778\times 10^{-3}$N.m/(rad/s)

L=4.38 mH

R=4.8 ohm

K=0.5

So the Transfer Function of the system

$$\frac{\omega}{Vin} = \frac{1}{\frac{R}{Kt}(S^2 Jeq + DeqS) + SKb} = \frac{1}{\frac{4.8}{0.5}(8.166034\times10^{-3}S^2 + 8.89778\times10^{-3}S) + 0.5S}$$

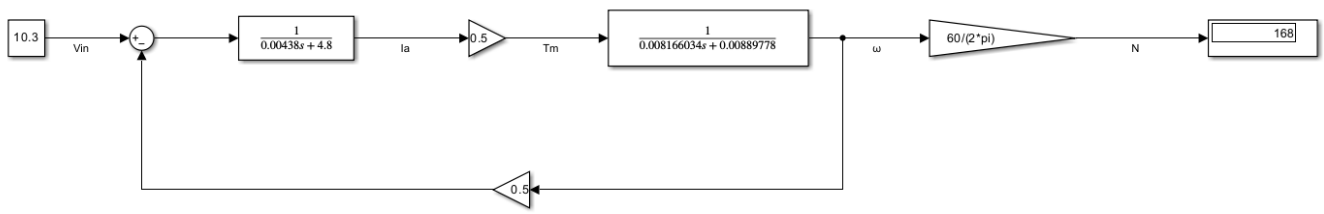



*Figure 24 mathematical system*

| Vin | N Real System | N mathematical System | Error% |
|---|---|---|---|
| 10.7 | 177 | 174.5 | 1.41 |
| 8.85 | 147 | 144.2 | 1.905 |
| 7.3 | 121 | 119.2 | 1.653 |
| 4.458 | 71.37 | 72.72 | 1.9 |
| 2 | 31.3 | 32.6 | 4 |

*Figure 25 parameter Estimation Error*

## Comment

- We select the signal pwm volt with 5v(255) because it is more suitable, stable and lower in error
- As we take the readings of the volt and rpm from a scope there will be error in readings
- As the AVO has an error in current measuring there will be an error
- Inertia of the motor is high because of the inertia of Gear box

## Drive Links:-

1- Cad simulation

https://drive.google.com/file/d/1fQp9lyu9A4hG4bAyTZ-pTlwYIAIv0jPN/view?usp=sharing

2- Cad Files

https://drive.google.com/file/d/1vqIbr92-q8fxsMk1YJDZMLcG7s7TOWOm/view?usp=sharing

3- MATLAB Files

https://drive.google.com/file/d/1cioaiNM_N_hUcws6125PWbWA_I5-uxSz/view?usp=sharing